Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Large language models now answer hard scientific questions, yet they still hallucinate and follow broken lines of reasoning. Multi-agent debate offers one way to push back on this. When several models work through a question before committing to an answer, the reasoning usually gets sharper and errors get caught. The weakness is in how these debates settle. Most systems pick a winner by majority vote or by which agent sounds most confident. When a confident majority is simply wrong, it can pull a correct minority agent off its answer, and the group lands on the wrong conclusion. No current method checks whether an agent's claims actually hold up against outside scientific evidence before letting that agent sway the room. The framework addresses this problem by adjusting how much influence each agent carries as the debate unfolds. It breaks every agent response into small factual claims, retrieves relevant scientific literature for each claim, and checks whether the evidence backs the claim or contradicts it. Every agent then carries a trust score that moves up or down as these checks accumulate over the rounds of debate. The trust update assigns less influence to unsupported claims, allowing a correct minority agent to retain influence even when outnumbered. The final answer is built from trust-weighted aggregation rather than a raw count of votes. The framework is expected to reduce hallucinations and keep reasoning tied to verifiable sources. It may also reduce premature agreement on a wrong answer while operating without model fine-tuning.

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Publication List

The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

1. Leveraging Machine Learning in Balancing Inventories and Analyzing Market Trends: A Comprehensive Systematic Review (Artificial Intelligence Review, Under processing)

Conference Papers

1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

1.

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Chapter 1

Introduction

The chapter introduces the study. It presents the background and problem, the motivation behind the work, the objectives, a short summary of the methodology, the expected outcome, and the structure of the rest of the report.

1.1 Project Overview

Large language models now handle many reasoning tasks, from scientific question answering to tutoring and decision support. A single model can still be confidently wrong. It may state an incorrect answer in fluent and assured language, which makes the mistake hard to catch. Multi-agent debate was proposed to reduce this problem. Several agents answer the same question, read each other's reasoning, and revise their positions over a few rounds before a final answer is selected [1]. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2], which makes the reasoning inside a debate easier to inspect. The real difficulty is the final step, where the separate answers are combined. Most debate systems settle disagreements by majority vote. Each agent receives the same weight, so the most common answer wins. This rule treats popularity as a stand-in for correctness. In scientific reasoning, a better answer should be backed by evidence rather than repeated by more agents. The study examines a trust-calibrated multi-agent deliberation framework that adjusts each agent's influence during the debate based on how well its claims match retrieved external evidence. The final answer then depends on evidence-grounded trust rather than a raw headcount.

1.2 Motivation

Majority voting can fail in a way that is easy to miss. A debate may look like it reached agreement while it has quietly dropped a correct answer held by one agent. Recent studies of multi-agent debate report that agents often copy or switch answers under social pressure instead of reasoning on their own, and that the correct answer is present in the discussion but ignored in more than 20 percent of wrong-answer cases [3]. Work on model behaviour points to a related tendency, where models trained to be agreeable grow more sycophantic and less accurate [4]. When a confident majority pushes a correct minority agent to give up its answer, majority voting does more than pick the wrong result. It hides the failure behind an appearance of consensus. This matters most for scientific questions, where a wrong but confident consensus can mislead a user into trusting an answer that deserved a closer check. Existing methods approach this failure from different directions. iMAD decides when a debate is worth running [5], DebUnc uses self-reported confidence to adjust agent influence [6], and Mixture-of-Agents combines outputs through static aggregation [7]. These methods are useful, yet none ties an agent's influence to whether its claims hold up against external evidence while the debate is still active. That gap is what motivates the study.

1.3 Objectives

The main objective is to reduce sycophantic consensus in multi-agent LLM debate by replacing plain majority voting with evidence-grounded trust calibration. This requires examining how a confident wrong majority can pressure a correct minority agent into changing its answer. It also involves designing a bounded trust score that rises when an agent's claims are supported by external evidence and falls when they are contradicted, then applying that score before the final answer is fixed. The study compares the method with standard majority voting and other relevant debate and aggregation baselines, and evaluates it using answer accuracy together with consensus-collapse, minority-preservation, and evidence-calibration metrics.

1.4 Methodology

The study follows an experimental methodology. A baseline multi-agent debate system is built first, where several heterogeneous LLM agents answer the same scientific question, share their reasoning, and revise across rounds. The baseline uses majority voting so the usual failure mode can be observed and measured. Each agent's reasoning is then broken into smaller factual claims. These claims are checked against scientific evidence retrieved from external sources such as academic papers and reference databases. Any contradiction between a claim and the retrieved evidence is detected and scored [8], building on the idea that external tools can verify and correct model output [9]. The verdicts are turned into trust updates. Supported claims raise an agent's trust, contradicted claims lower it, and uncertain claims are handled conservatively. The trust score is bounded so that no agent gains unlimited influence or drops out of the debate completely. The final answer is chosen through trust-weighted aggregation rather than a simple count. The framework is evaluated on scientific reasoning datasets such as GPQA and MMLU-Pro. The evaluation checks whether the mechanism lowers consensus collapse, keeps correct minority answers, and holds overall accuracy without any model fine-tuning.

1.5 Project Outcome

The main outcome is a reproducible framework for trust-calibrated multi-agent scientific deliberation. It shows how retrieved evidence can update agent influence during a debate, and how the final answer can be decided by evidence-weighted trust instead of a majority vote. Alongside it, the study produces an evaluation protocol that detects sycophantic consensus, measures how often correct minority answers survive, and compares the method against established baselines. The study is expected to show whether dynamic trust calibration makes multi-agent debate more dependable on scientific questions. Even where the method does not remove every wrong answer, it should make the decision more transparent by recording which agents were supported or contradicted by the evidence.

1.6 Organization of the Report

The rest of the report is arranged as follows. Chapter 2 covers the background, the related work on multi-agent debate and sycophancy, and the gap addressed by the study. Chapter 3 presents the requirement analysis, the detailed methodology, the system design, and the project plan. Chapter 4 describes the implementation setup, the testing and evaluation procedure, and the results with discussion. Chapter 5 discusses the relevant standards, the impact of the work, sustainability, and the complex engineering problem analysis. Chapter 6 concludes the report, notes its limitations, and gives recommendations for future work.

Chapter 2

Background

The chapter introduces the concepts needed to understand the system described in the report and reviews research on multi-agent debate, sycophancy, and agent influence. It then explains the gap addressed by the study.

2.1 Preliminaries

Large language models (LLMs) generate answers one token at a time, but their outputs can still contain unsupported or incorrect reasoning. Fluent wording does not guarantee that an answer is correct [10]. Chain-of-thought prompting asks a model to expose intermediate reasoning steps before giving a final answer, which makes the reasoning easier to inspect and gives later components more information to evaluate [2]. These steps are useful evidence about how an answer was formed, but they should not be treated as proof of correctness by themselves. Multi-agent debate extends this process by asking several agents to solve the same problem independently. The agents then receive the other answers and explanations, discuss their differences, and revise their positions over a fixed number of rounds. A final answer is selected after the discussion [1]. Several agents are useful because their independent attempts can expose different errors and allow one agent to question another agent's reasoning. The way those answers are combined still matters. A common method is majority voting, where each agent has the same weight and the most frequent answer is selected. If several agents repeat the same mistake, the vote can strengthen the mistake instead of correcting it [11]. Sycophantic consensus refers to an agent copying or abandoning a correct position because a wrong position has greater support [3]. Confidence may be supplied by an agent or estimated from its token distribution, but neither signal directly verifies a claim [6]. DebUnc shows this difference by comparing deployable uncertainty measures with a diagnostic Ground Truth measure that uses the correct answer [6]. The framework therefore defines trust through support from retrieved scientific evidence.

2.2 Literature Review

The review covers studies that address debate construction, external verification, sycophancy, uncertainty, and debate evaluation. The main comparison is between methods that decide when to debate, methods that change agent interaction, and the method that changes agent influence using retrieved evidence. Recent advances in reasoning-oriented LLMs have shown that improving reasoning quality requires more than increasing the size of the model. Research has increasingly focused on structured reasoning, multi-agent collaboration, and external verification to improve the reliability of generated responses [10]. Multi-agent debate has emerged as one of the most widely studied approaches because it enables agents to challenge each other's reasoning before producing a final answer [1]. The following review examines representative studies in these areas and identifies the remaining research gap addressed by the proposed framework.

2.2.1 Similar Applications

The system is not a conventional web or mobile application. Instead, it is an experimental deliberation layer that can be integrated into scientific question-answering systems, research assistants, evidence-based decision-support tools, or other applications that require reliable reasoning. Rather than relying on a single model response, these systems generate multiple candidate solutions, allow the candidates to interact, and aggregate the results before returning a final answer to the user. Du et al. demonstrated that structured debate among multiple agents can improve factuality and reasoning compared with a single-agent approach [1]. Wang et al. further showed that combining the outputs of several models through layered aggregation can improve overall model capability without requiring explicit debate [7]. Existing multi-agent applications differ mainly in how they coordinate agents and combine their outputs. Some systems focus on structured debate to encourage reasoning refinement between agents [1]. Other systems organize multiple agents into hierarchical aggregation pipelines where later models refine or combine earlier responses [7]. Zhou et al. proposed a self-organizing architecture that dynamically selects reasoning agents according to the task being solved rather than using a fixed workflow [12]. Kushwaha and Madhavan introduced a retrieval-augmented multi-agent system that identifies contradictions among retrieved documents and ranks evidence according to source credibility before generating a final response [8]. These studies demonstrate that distributing reasoning across multiple specialized agents can improve robustness and factual consistency, although their aggregation strategies remain largely fixed once deliberation begins. The proposed framework shares the objective of improving answer reliability but differs in how agent influence evolves during discussion. Rather than assigning equal influence throughout the debate or relying only on confidence estimates, the framework records evidence verification results after every debate round and continuously updates each agent's trust score according to claim-level scientific evidence. This enables the system to preserve well-supported minority opinions while reducing the influence of unsupported majority consensus.

2.2.2 Related Research

Wei et al. (2022) showed that chain-of-thought prompting can improve reasoning in large language models [2]. Du et al. (2024) extended this idea through multi-agent debate, where agents answer a question, examine peer reasoning, and revise their positions [1]. Gou et al. (2024) proposed CRITIC, which uses external tools to check and improve model responses [9]. The method supports external verification, but it does not use several models debating the same question. Wang et al. (2025) introduced Mixture-of-Agents, a layered method that combines outputs from several proposer models [7]. It shows the value of multiple LLMs, but it uses static aggregation without an explicit evidence-based trust value. Fan, Yoon, and Ji (2026) proposed iMAD to reduce the cost of running debate on every question [5]. Their method extracts 41 features from a structured self-critique and uses a classifier to decide whether debate should start. It reports token savings of up to 92 percent compared with GroupDebate and an accuracy improvement of up to 13.5 percentage points over a single-agent baseline. It does not change agent influence after debate begins. Pitre, Ramakrishnan, and Wang (2025) studied sycophancy inside multi-agent debate [3]. Their work found that the correct answer can be present but ignored in more than 20 percent of wrong-answer cases. CONSENSAGENT detects stalling and answer copying, then rewrites the task prompt before another debate stage. It reduces measured sycophancy, but its final aggregation still uses self-reported confidence and consistency. Yoffe, Amayuelas, and Wang (2025) proposed DebUnc, which shares agent uncertainty through prompts or attention scaling during debate [6]. Its deployable uncertainty measures give small gains, while a diagnostic Ground Truth signal gives a much larger gain. The result shows that the quality of the trust signal is a major limitation. Kushwaha and Madhavan (2026) described an eight-agent MAS-RAG architecture that detects contradictions among retrieved documents and ranks evidence by credibility [8]. Its source credibility is fixed by a human curator and is not updated during deliberation. Cau et al. (2025) examined opinion convergence in LLM interactions and found that selective persuasion can influence convergence [13]. Yeow et al. (2025) compared several decision architectures and reported that feedback between agents can allow early errors to spread [14]. Wang et al. (2024) found that a single LLM with strong in-context examples can sometimes match multi-agent discussion systems, and they identified judge errors and peer conformity as weaknesses [11]. Choi, Zhu, and Li (2025) separated debate communication from voting and argued that majority voting accounts for much of the usual improvement [15]. Cui et al. (2026) proposed FREE-MAD, which avoids final consensus and uses anti-conformity prompts with trajectory scoring [16]. Zhou et al. (2025) proposed ReSo, a reward-driven system that selects agents through dynamic task graphs [12]. These works study coordination and agent selection, but they do not verify scientific claims before changing agent influence. Ibrahim et al. (2026) reported that tuning models to sound warm and agreeable can reduce factual accuracy and increase sycophancy [4]. Pitre et al. (2026) proposed process-level diagnostics for engagement, responsiveness, influence asymmetry, balance, stability, and agent utility in debates where ground truth may be unavailable [17]. Their work shows that final agreement alone is not enough to judge a debate. These studies address debate construction, external checking, uncertainty sharing, prompt correction, and process evaluation. None continuously updates an agent's influence according to claim-level support from retrieved scientific evidence. The framework addresses that gap with a bounded trust score.

2.3 Gap Analysis

The reviewed literature reveals three remaining needs in multi-agent debate systems. First, a system should determine whether the additional computational cost of the debate is justified. Fan et al. addressed this through iMAD by predicting whether a debate is necessary before it begins [5]. Second, a system should detect unsupported agreement caused by conformity or sycophancy. CONSENSAGENT mitigates this problem through prompt optimization during debate [3]. Third, a system should determine which agents deserve greater influence when disagreement remains. DebUnc introduces uncertainty-based weighting, but its trust signals rely on internal model uncertainty rather than independently verifiable evidence [6]. Previous studies also show that agreement does not necessarily indicate correctness. Cau et al. found that selective persuasion influences opinion convergence [13]. Wang et al. identified peer conformity as a limitation of multi-agent debate [11]. Existing retrieval-based systems verify retrieved information or improve aggregation, but they do not continuously update an agent's influence according to verified scientific evidence during debate [9]. Dynamic coordination methods also focus on agent selection rather than evidence-driven trust updates [12]. The remaining gap is therefore an evidence-grounded mechanism that continuously updates agent influence during an ongoing debate. The proposed framework addresses this gap by decomposing an agent's response into factual claims, retrieving relevant scientific passages, and classifying each claim as supported, contradicted, or unverifiable. These verification results update a bounded trust score after every debate round, allowing well-supported minority agents to retain influence while reducing the impact of unsupported majority consensus. The framework is evaluated using consensus-collapse rate, minority-preservation rate, and evidence-calibration rate alongside conventional answer accuracy. Retrieval sparsity, imperfect claim decomposition, and correlated reasoning errors between agents remain limitations of the proposed approach.

2.4 Summary

The chapter introduced LLM reasoning, chain-of-thought prompting, multi-agent debate, and sycophantic consensus. The reviewed studies cover debate efficiency, prompt correction, uncertainty weighting, external verification, process evaluation, and agent coordination. They do not continuously update agent influence from claim-level scientific evidence. The next chapter uses this gap to define the system requirements, detailed methodology, and project plan.

Chapter 3

Project Design

This chapter defines the system requirements, presents the detailed methodology and the design decisions behind the framework, and lays out the project plan with the task allocation for the team. The first section records the functional and nonfunctional requirements together with the context diagram, the level-one data flow diagram, and the interface design.

3.1 Requirement Analysis

Requirement analysis fixes what the framework must do before any code is written. The system is an experimental deliberation layer rather than a conventional application, so the requirements are framed around the research pipeline. A question enters, agents debate it over a fixed number of rounds, the claims are checked against external evidence, and a trust-weighted answer comes out. Multi-agent debate was chosen because it has been shown to improve factuality over single-model answers [1]. The trust mechanism exists because a confident wrong majority can push a correct minority agent into abandoning its answer [3]. The subsections below define the requirements, the system boundary, the data flows, and the interface through which the framework is operated.

3.1.1 Functional and Nonfunctional Requirements

The functional requirements describe the services the framework must provide. The system accepts a scientific question from a user or the evaluation harness and keeps the ground truth hidden from all agents. For every question the confidence gate decides whether a debate is needed, sends easy questions to a direct answer, and routes the remaining questions into the debate pipeline. The orchestration process runs three heterogeneous LLM agents over at most three revision rounds, with a retry cap of three, and sequences the rounds with a state machine. Each agent's reasoning is split into atomic factual claims tagged to their source agent, with a fallback extraction pass when structured tagging fails. The retrieval subsystem searches each agent's claims in a different database, PubMed for the first agent, arXiv for the second, and Semantic Scholar for the third, with

OpenAlex as a fallback when a source is unavailable. Retrieved passages are reranked with a cross-encoder and every claim receives a verdict of supported, contradicted, unverifiable, or contested. After every round the trust score of each agent is updated with $S_i(t+1) = S_i(t) + \alpha V_i - \beta H_i$, then softmax is applied, the values are clamped between 0.1 and 0.9, and the vector is renormalized so that no agent is silenced or dominant. The system also supports an injection point where a fabricated wrong consensus can be inserted between rounds, so that sycophantic collapse can be measured under controlled pressure. The final answer is selected by weighted aggregation of trust and position at the last round rather than by majority vote, and the result package contains the answer, the supporting citations, the full trust trajectory, and each agent's reasoning trace. The evaluation harness computes answer accuracy, consensus-collapse rate, minority-preservation rate, and evidence-calibration rate, and compares the framework with majority voting and with the MoA [7], iMAD [5], and DebUnc [6] baselines. Finally, the system recovers unparseable model output with a fallback extraction pass, retries a timeout once before marking the round inconclusive, and falls back to cached evidence or OpenAlex when a retrieval API is down.

The nonfunctional requirements describe the qualities of the system rather than its functions. The debate must respect the round cap of three and the retry cap of three so that latency stays bounded on the target hardware. A failure in one component must not block the whole debate, and every failure path must degrade locally and keep the pipeline moving. The framework must work with closed APIs and locally served models through the OpenAI-compatible vLLM interface, and the pipeline must run unchanged on the development models (Qwen3.5-9B, Gemma 4 12B, Ministral-3-14B) and on the final models (Qwen3.6-27B, Gemma 4 26B, Mistral Small 3.2 24B). Inference must fit within a single A100 80GB rental at about USD 0.68 to 1.50 per hour, with no fine-tuning. Every run must be seeded and every intermediate result logged so that any experiment can be repeated and checked. The output must record which agent was supported or contradicted by the evidence so that the decision trail can be audited. The ground truth must never be exposed to the agents, and the final output must cite the evidence behind the answer.

3.1.2 Context Diagram

The context diagram in Figure 3.1 shows the system boundary and the external actors it exchanges data with. The framework sits in the middle of the diagram. It receives questions from the user or the evaluation harness, retrieves scientific passages from three external databases, and calls off-the-shelf language models through the vLLM server running on a rented A100 GPU. The evaluation harness receives the scores produced by the framework, and the user receives the final result package with the answer, the evidence, and the trust trajectory.

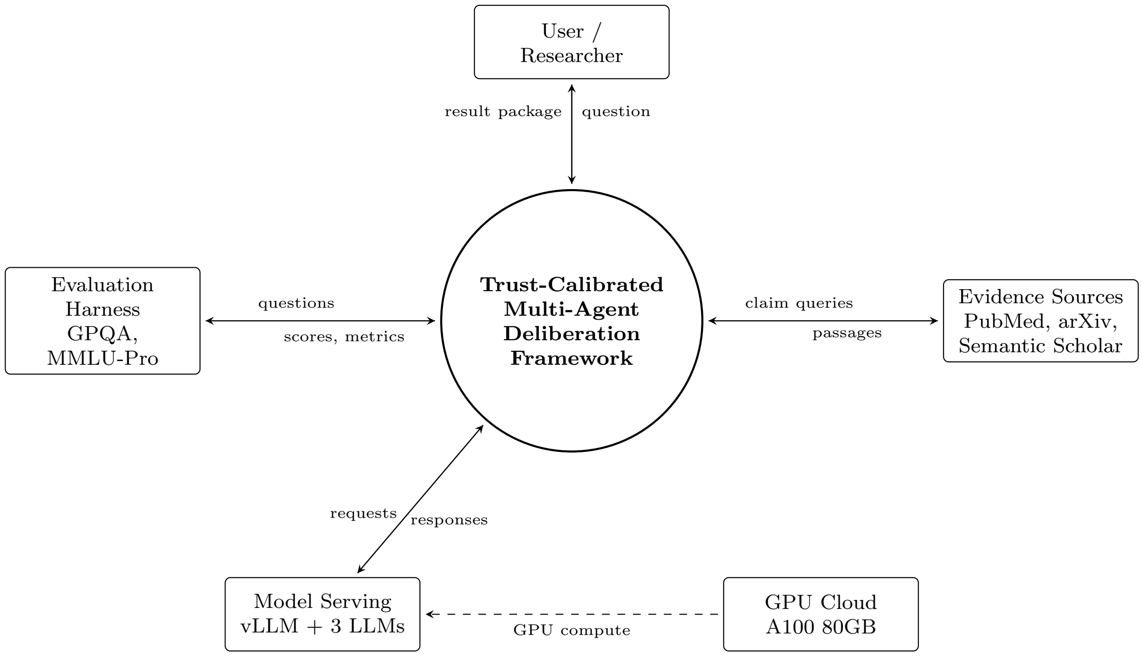

Figure 3.1: Context diagram of the trust-calibrated multi-agent deliberation framework.

3.1.3 Data Flow Diagram Level 1

The level-one data flow diagram in Figure 3.2 decomposes the framework into seven processes. A question enters through the intake and gate process, which decides between a direct answer and a debate. The orchestration process runs the revision rounds, sends inference requests to the external models, and reads and writes the debate state. Each position is decomposed into atomic claims and verified against the external sources. The verdicts drive the trust update, which writes the new trust values back into the debate state. At the last round the weighted aggregation process selects the answer, and the packaging process logs the result and returns it to the user. Three data stores hold the debate state, the evidence and verdict log, and the results log.

The diagram was drawn with Mermaid. The code below can be pasted directly into mermaid.live to regenerate it, and the exported image is placed in Figure 3.2.

```
flowchart LR
    EH["Evaluation Harness<br/>GPQA, MMLU-Pro"]
    US["User / Researcher"]
    ES["Evidence Sources<br/>PubMed, arXiv,<br/>Semantic Scholar"]
    MS["Model Serving<br/>vLLM + 3 LLMs"]

    P1(("P1 Intake and Gate"))
    P2(("P2 Debate Orchestration"))
    P3(("P3 Claim Decomposition"))
    P4(("P4 Retrieval and Verification"))
```

```
P5(("P5 Trust Update"))
P6(("P6 Weighted Aggregation"))
P7(("P7 Result Packaging"))

D1[("D1 Debate State")]
D2[("D2 Evidence and Verdict Log")]
D3[("D3 Results Log")]

EH -->|question| P1
P1 -->|debate = yes| P2
P2 -->|positions| P3
P3 -->|atomic claims| P4
P4 -->|queries| ES
ES -->|passages| P4
P4 -->|verdicts| P5
P4 -->|verdicts| D2
P5 -->|trust updates| P6
P5 -->|trust| D1
P2 -->|positions| D1
D1 -->|state| P2
P6 -->|answer| P7
P7 -->|result package| US
P7 -->|log| D3
P2 -->|inference requests| MS
MS -->|responses| P2
```

Generated data flow diagram (paste the code above into mermaid.live, export the image, and place it here).

Figure 3.2: Level-one data flow diagram of the trust-calibrated multi-agent deliberation framework.

3.1.4 UI Design

The framework is operated through a command-line interface and a small web dashboard. The command-line interface starts experiments, points the pipeline at the evaluation datasets, and writes the result of every run to a JSON package. It is the primary interface because the system runs in batches on a rented GPU, where a graphical interface adds little value. The web dashboard is used afterwards to inspect single debates. It shows the agents as cards with their current positions, a chart of the trust trajectory across rounds, and the verdict of every claim with the passage that supports or contradicts it. This is where the transparency of the framework becomes visible, because a user can trace exactly why an agent lost influence. Figure 3.3 and Figure 3.4 show the intended layout of both interfaces. The screenshots will be replaced with the real output of the system once the implementation is complete.

Screenshot of the command-line experiment interface.

Figure 3.3: Command-line interface for running experiments and writing JSON result packages.

Screenshot of the debate inspection dashboard.

Figure 3.4: Web dashboard showing agent positions, the trust trajectory, and per-claim evidence verdicts.

3.2 Detailed Methodology and Design

You have to mention alternate solutions that you have considered. Why you have selected the specific solution, etc.

3.3 Project Plan

3.4 Task Allocation

3.5 Summary

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only include standards related to the system. This list is not exhaustive. For each standard, discuss the alternatives, their advantages and disadvantages, and the reason for selecting it.

5.1.1 Software Standards

5.1.2 Hardware Standards

5.1.3 Communication Standards

5.2 Design Constraints

Only include constraints related to the design. This list is not exhaustive.

5.2.1 Economic Constraint

5.2.2 Environmental Constraint

5.2.3 Ethical Constraint

5.2.4 Health and Safety Constraint

5.2.5 Social Constraint

5.2.6 Political Constraint

5.2.7 Sustainability

5.3 Cost Analysis

Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with complex problem solving.

P1 Dept of Knowledge	P2 Range of Conflicting Requirements	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stakeholder Involvement	P7 Interdependence
√	√					

5.4.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.2).

Table 5.2: Mapping with complex engineering activities.

A1 Range of resources	A2 Level of Interaction	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
√	√			

5.5 Summary

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

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